

SHELDON G.B. WAUGH, MSC, PHD

939 Ramble Run Rd, Middle River, MD, 21220

waughsh@gmail.com ♦ <https://waughr.us> ♦ <https://www.linkedin.com/in/waughsh> ♦ (239)404-8668

BASIC INFORMATION

Citizenship

Yes - United States and United Kingdom

Federal Experience

June 2018 - Present

United States Army Medical Command, Army Public Health Center

United States Census Bureau, Department of Commerce

United States Department of Commerce, Office of Inspector General

Military Experience

United States Army Reserve, Signal Corps, Major

December 2011 - Present

EDUCATION

University of Florida, Gainesville, Florida

August 2007 - May 2018

Department of Epidemiology, College of Public Health and Health Professions

August 2014 - May 2018

PhD in Epidemiology

Department of Geography, College of Liberal Arts and Sciences

August 2007 - May 2014

Masters of Science in Geography

Bachelors of Science in Geography

WORK EXPERIENCE

Data Analytics Branch, U.S. Department of Commerce, Office of Inspector General June 2023 - Present

Lead Data Scientist GS-1560-14

Supervisor: Andrew Brown

- Chief and principal data scientist for the DOC OIG.
- Lead and managed the Data Analytics team in conducting data science research and projects that make informed projections, data-driven decisions, and recommendations based on findings.
- Developed, maintained and administered an ArcGIS Online-based mapping and analysis dashboard to track broadband availability and progress, supporting the Department of Commerce Office of Inspector General's mission to ensure efficient and effective broadband infrastructure deployment. Utilized Python and data from the FCC and NTIA to identify unserved and underserved areas, monitoring the impact of federal funding, and visualizing key metrics for stakeholders.
- Participated in the CIGIE and PRAC data sharing pilot using Entity recognition and a unified Databricks platform for enterprise-grade analytics and AI solutions. Collaborated with AI partners to integrate Machine Learning technologies, increasing lead generation efficiency.
- Developed and maintained 3 ArcGIS Online dashboards for DOC OIG, covering PRAC PPP loan usage, NOAA hurricane errors, and IJJA broadband deployment. Created an advanced script for web scraping and API usage to gather geospatial metadata from the Department of Commerce for the Geospatial Data Act, processing 30 GB of data using R and Python. Works as administrator of DOC OIG's ArcGIS Enterprise's Server environment.
- Developed and recommended the Data Governance Guidelines for the Department of Commerce, Office of the Inspector General (DOC OIG). The DOC OIG Data Analytics Branch developed these guidelines to provide a foundation of best practices for curating data from server-based databases for reports, visualizations, and analyses.
- Developed a proposal to the GSA Technology Modernization Fund to implement Natural Language Processing on large corpora of grant documentation and unstructured text to identify potential pass-through schemes. Additionally, proposed utilizing generative AI to create an internal chatbot, summarizing large amounts of state broadband deployment documents and files into manageable summaries for investigators.

Field Quality Monitoring Branch, Office of Survey and Census Analytics, Field Division, US Census Bureau January 2021 - June 2023

Data Scientist GS-1560-13

Supervisor: Mary Davis

- Provided expertise in the applications of data science (interdisciplinary analytical, statistical, and programming skills) to develop data-driven solutions for complex business challenges.

- Led an audio-related natural language processing project in collaboration with the field of social, economic, and housing statistics division. Coordinated access to Computer Audio Recorded Interview (CARI) Interactive Data Access (CIDA) system to download multiple years of SIPP interview data. Successfully submitted and approved Python package Transformers through the Standards Working Group for access to open-source machine learning models through HuggingFace, an AI Repository.
- Led, established and administered the development of front-facing ESRI Arc GIS Enterprise Mapper dashboards, maps, and tools for the American Housing Survey (AHS) and the National Health Interview Survey (NHIS), an initial and essential part of the Field Quality Monitoring Program, stood up by the Office of Census Analytics (OSCA). Worked as administrator of OSCA's ArcGIS Enterprise's Server environment.
- Led and established the development of front-facing Tableau dashboards for the American Housing Survey (AHS), an initial and essential part of the Field Quality Monitoring Program (FQM), stood by OSCA. Worked as senior administrator of OSCA's Tableau Server, a cloud-based server.
- Led the development of models for outlier and anomaly detection for Field Representatives and Field Supervisor Areas using AHS INR data. Researched and utilized novel coding methodologies using Python, pyspark and R and integrated them within the US Census Bureau survey database infrastructure.
- Developed extensive documentation of outlier/anomaly detection and metric development in Jupyter Notebooks, providing a clear and robust work trail from data extraction, transformation, analysis, and visualization.
- Developed and managed a Division-level Red Hat Enterprise Linux server environment to automate queries, metric calculations, data uploads, and visualizations. Served as the SAS Server Administrator for the division, responsible for maintaining the server with updates and hotfixes.

One Health Division, Veterinary Service and Public Health Sanitation Directorate, Army Public Health Center, Aberdeen Proving Ground June 2018 - January 2021

Epidemiologist GS-0601-11/12/13

Supervisor: LTC Sara Mullaney DVM, PhD, DACVPM

- Serves as the technical expert and advisor in the Veterinary One Health Division within the Veterinary Services and Public Health Sanitation Directorate, U.S. Army Public Health Center.
- Principal Investigator of a 350K grant focused on integrating high-performance computing, machine learning, and AI in Army Veterinary Services to enhance surveillance of companion animal diseases. Collaborated with Stanford University to use natural language processing and neural networks for analyzing veterinary records and applying diagnostic codes. The project aimed to improve surveillance capabilities and establish AI's role in public health within the DOD. Developed scripts in R and Python to process over 3,000 electronic clinical notes, with plans to expand to over 30 GB of DOD data, while navigating HealthIT governance for DOD EHR data.
- Plans and executes epidemiological projects and outbreak investigations aimed at identifying population based risk factors for zoonotic disease, infections and/or acute or chronic illnesses, injuries, and occupational hazards.

Spatial Epidemiology & Ecology Research Laboratory (SEER), Department of Geography, University of Florida March 2017 - June 2018

Bioinformatician

PI: Jason K Blackburn, PhD

- Chief Bioinformatics analyst and developer in genomic and spatial analysis
- Utilized advanced statistical and phylogenetic techniques to update genomic laboratory procedures in order to standardize lab results

Department of Epidemiology, College of Public Health and Health Professions, University of Florida August 2014 - June 2018

Research/Study Coordinator

PI: Volker Mai, PhD

- Bioinformatics analyst and developer in metagenomics
- Responsible for creation of data algorithms and specialized software pipelines to identify and classify components of 16S sequencing fragments.

Department of Geography, College of Liberal Arts and Sciences, University of Florida January 2013 - July 2013

Research/Data Analyst

PI: Andrew Tatem, PhD

- Tasked with data entry and processing of country population data to GIS databases
- Data used in AfriPOP and AsiaPOP projects with an aim of producing population distribution maps

TEACHING AND MENTORING EXPERIENCE

Science and Mathematics Academy, Aberdeen High School, Aberdeen, MD

June 2019- 2020

Mentor: Thomas Carey

Program Supervisor: Sarah Ashley

- Assists in project development, assessing the burden of antibiotic prescriptions among our companion animal population
- Provides research, logistical and overall support to the mentee in developing a fully-fledged and validated project
- Provides a synergistic relationship between mentor and mentee to allow for a collaborative and cooperative experience.

Department of Epidemiology, College of Public Health and Health Professions, University of Florida

January – May 2015

Teaching Assistant (PHC6003: Epidemiology of Chronic Diseases)

Instructor: David Sheps MD, MSPH

- Assisting Instructor with developing, grading assignments quizzes and exams
- Provided presentations and online lectures to students

Department of Epidemiology, College of Public Health and Health Professions, University of Florida

January – May 2015

Teaching Assistant (PHC4101: Public Health Concepts)

Instructor: Sarah McKune MPH, PhD

- Assisting Instructor with developing, grading assignments quizzes and exams
- Provided presentations and online lectures to students

RESEARCH SUPPORT

Accelerating Innovation in Military Medicine Research - **DM190430** - Integrating High-Performance Computing and Machine Learning Within the Army Veterinary Service to Improve Surveillance of Companion Animal Disease Within the Department of Defense

March 2020 - January 2022: 280,000

The project will use ML and Natural Language Processing (NLP) models and algorithms to annotate, code and categorize unstructured text in the form of veterinary encounter notes (eNotes) with SNOMED (ID-10-like codes for Animals) codes and categorizations. Our approach will include structured categorization that will improve the current state of veterinary surveillance by harnessing the ability to access the robust data source of eNotes and the diagnoses and observations within. From this output, for the companion animal population that includes military working dogs and service members' pets seen around the world at veterinary treatment facilities (VTF), our objective is to accurately and timely assess the current burden of disease and identify risk factors for disease.

Role: Principal Investigator

TECHNICAL STRENGTHS

Advanced Biostatistical Methods

Machine Learning Methods

Geo-spatial Modeling and Analysis

Data Management and Cleaning Methods

Data Visualization Solutions

Software & Tools

Machine Learning and Bayesian modeling

NLP and Advanced Regression techniques

ESRI ArcGIS Enterprise, Database management

SQL, R and, NoSQL coding techniques

Tableau, Power BI, R, Python

Python, R, VB, linux, bash and C++ (12+ years of proficiency)

HONORS, ACHIEVEMENTS AND AWARDS

40 Gators Under 40/Outstanding Young Alumni Award, University of Florida, Gainesville, Florida *April 2025*

Department of Commerce Bronze Award, U.S. Census Bureau, Suitland, Maryland *August 2023*

Director's Award for Innovation, U.S. Census Bureau, Suitland, Maryland *August 2022*

Army Commendation Medal, U.S. Army Reserve, Chicopee, Massachusetts *July 2022*

Individual Cash Perf Award, U.S. Census Bureau, Suitland, Maryland *May 2022*

Army Medicine Annual Wolf Pack Award (FY21), U.S. Army Public Health Center (APHC) COVID-19 Task Force Team, AMEDD Civilian Corps *April 2022*

Group Special Act Performance Award, U.S. Census Bureau, Suitland, Maryland *September 2021*

Army Commendation Medal, U.S. Army Reserve, Milton, Florida *October 2019*

Honorable Mention, Student Research Abstract Award, SHES/APHA Annual Meeting *November 2016*

SMART Scholarship, Department of Defense, Washington D.C. *August 2016 - May 2018*

McKnight Fellowship, Florida Education Fund, Orlando, Florida *August 2014 - May 2018*

Ryan Poehling Fellowship Award, University of Florida	<i>December 2013 - May 2014</i>
Army Achievement Medal, U.S Army Reserve, Milton, Florida	<i>June 2014</i>
Army Achievement Medal, U.S Army Reserve, Milton, Florida	<i>December 2013</i>
LTC Samuel W Anderson Scholarship, University of Florida	<i>December 2009 - May 2011</i>
1LT Mark T Barrett Memorial Award, University of Florida	<i>May 2009 - May 2010</i>
Gold Scholarship, University of Florida	<i>August 2007</i>

PROFESSIONAL SOCIETIES AND ORGANIZATIONS

The National Association of County Health Officials	<i>March 2019 - Present</i>
International Society of Disease Surveillance	<i>August 2018 - Present</i>
Association for Veterinary Informatics (Education Committee)	<i>July 2018 - Present</i>
AMSUS - The Society of Federal Health Professionals	<i>May 2018 - Present</i>
Association of American Geographers	<i>September 2013 - Present</i>
American Public Health Association (Treasurer - One Health Group)	<i>October 2014 - Present</i>

MILITARY EXPERIENCE

Headquarters Company, 302nd Maneuver Enhancement Brigade, Chicopee, Massachusetts, United States Army Reserve
Major, Signal Corps

- Brigade Communications Officer *August 2020 - Present*
- Network Operations Officer *September 2018 - July 2020*

842nd Signal Company, Milton, Florida, United States Army Reserve
Captain, Signal Corps

- Company Commander *September 2015 - 2018*
- Family Readiness Group Liaison *January 2015 - 2018*
- Company Executive Officer *September 2013 - 2015*
- Platoon Leader *December 2011 - 2013*

PUBLISHED WORKS

Bayko, H.; Watkins, S.; **Waugh, S.**; Moore, G.; Mullaney, S. Adaptation of the One Health Zoonotic Disease Prioritization Tool for Government- and Privately Owned Companion Animal Zoonotic Disease Surveillance. *Zoonotic Dis.* 2023, 3, x. <https://doi.org/10.3390/zoonoticdis3030020>

Seal, L., Mullaney, S, **Waugh, S.** 2022. Leishmaniasis in the United States military veterinary patient population. *Journal of the American Veterinary Medical Association.* PMID: 34780354

Wijayabahu, A.T.,**Waugh, S.**, Ukhanova, M. and Mai, V., 2019. Dietary raisin intake has limited effect on gut microbiota composition in adult volunteers. *Nutrition journal*, 18(1), p.14. PMID: 30845997

Tagliamonte, M. S.,**Waugh, S.**, Prosperi, M., & Mai, V. (2019, September). An Integrated Approach for Efficient Multi-Omics Joint Analysis. In *Proceedings of the 10th ACM International Conference on Bioinformatics, Computational Biology and Health Informatics* (pp. 619-625). ACM. PMID: 31588431

Waugh, S., and Mullaney, S. "Progress towards Companion Animal Zoonotic Disease Surveillance in the US Army." *Online Journal of Public Health Informatics* 11.1 (2019).

Ball, J. D., Fe Agana, D., **Waugh, S.**, Wang, K., James, T. G., & Nicolette, G. (2019). Systematically collected information at encounters with HIV-positive students: A review of 10 years of electronic medical records. *Journal of American College Health*, 1-5. PMID: 30681932

Jennifer C. Dennis, Tyler Culpepper, Carmelo Nieves, Jr., Cassie C. Rowe, Alyssa M. Burns, Carley T. Rusch, Ashton Federico, Maria Ukhanova, **Waugh, S.**, Volker Mai, Mary C. Christman, Bobbi Langkamp-Henken, Probiotics (Lactobacillus gasseri KS-13, Bifidobacterium bifidum G9-1, and Bifidobacterium longum MM-2) improve rhinoconjunctivitis-specific quality of life in individuals with seasonal allergies: a double-blind, placebo-controlled, randomized trial. *Am J Clin Nutr* 105, 758–767 (2017). PMID: 28228426

Waugh, S. App.: Gut Microbiota Differences in Children From Distinct Socioeconomic Levels Living in the Same Urban Area in Brazil. *Journal of Pediatric Gastroenterology and Nutrition* (2016). PMID: 28644365

Oliveira, F.P. de, Mendes, R.H., Dobbler, P.T., Mai, V., Pyro, V.S., **Waugh, S.**, Vairo, F., Refosco, L.F., Roesch, L.F.W., and Schwartz, I.V.D. (2016). Phenylketonuria and Gut Microbiota: A Controlled Study Based on Next-Generation Sequencing. *PLOS ONE* 11, e0157513. PMID: 27336782

Dahl, W. J., Ford, A.L., Ukhanova, M., Radford, A., Christman, M.C., **Waugh, S.**, Mai, V. Resistant potato starches (type 4 RS) exhibit varying effects on laxation with and without phylum level changes in microbiota: A randomised trial in young adults. *Journal of Functional Foods* 23, 1–11 (2016).

Waugh, S. Apropos: *Plasmodium knowlesi* malaria an emerging public health problem in Hulu Selangor, Selangor, Malaysia (2009–2013): epidemiologic and entomologic analysis. *Parasites & Vectors* 8, 79 (2015). PMID: 25651916

Mai, V., **Waugh, S.**, Byrd, D., Simpson, D. & Ukhanova, M. Novel encapsulation improves recovery of probiotic strains in fecal samples of human volunteers. *Appl Microbiol Biotechnol* 1–7 (2016). PMID: 27796434

PRESENTED WORKS

Waugh, S., Developing a comparison metric for survey question compliance: A case in utilizing open-source software and methodologies within the US Census Bureau, 64th ISI World Statistics Congress - Ottawa, Canada, 2023

Waugh, S., Hollowood, M., Denby, R., The Evolution of the Field Quality Monitoring (FQM) Mapper, AAG Annual Meeting, 2023

Waugh, S., Integrating High-Performance Computing and Machine Learning within the Army Veterinary Service to improve Surveillance of Companion Animal Disease within the Department of Defense, AVI Talbot Symposium, 2020

Waugh, S., Progress Towards an Integrative and Data-Driven Companion Animal Zoonotic Disease Surveillance System within the Department of Defense, World One Health Congress, 2020

Waugh, S., Progress Towards an Integrative and Data-Driven Companion Animal Zoonotic Disease Surveillance System within the Department of Defense, Military Health System Research Symposium, 2020

Waugh, S., Progress Towards an Integrative and Data-Driven Companion Animal Zoonotic Disease Surveillance System within the Department of Defense, Consortium on Global Health Annual Meeting, 2020

Bayko, H, **Waugh, S.**, Watkins, S, Mullaney, S , Zoonotic Disease Prioritization for Government and Privately Owned Companion Animal Zoonotic Disease Surveillance System: Adaptation of the One Health Zoonotic Disease Prioritization Tool, American Public Health Association Annual Meeting, 2019

Waugh, S., Progress towards an Integrated Companion Animal Zoonotic Disease Surveillance System within the DoD, International Society of Disease Surveillance Annual Meeting, 2019

Waugh, S., Progress towards an Integrated Companion Animal Zoonotic Disease Surveillance System within the DoD, APHC Science Exchange, 2019

Waugh, S., Progress towards an Integrated Companion Animal Zoonotic Disease Surveillance System within the DoD, APHC One-Health Day Seminar, 2018

Waugh, S., Sytnik, I, Karibayev, T, Alimbayev, A, Ornybayev, M, Rametov, M, Nikolich, M, Hagius, S, Elzer, P, Blackburn, J. Brucellosis Transmission Between Humans and Domesticated Livestock In Southern Kazakhstan: Inferences Through MLVA Typing, UF Emerging Pathogens Institute Research Day, 2018

Waugh, S., Sytnik, I, Karibayev, T, Alimbayev, A, Ornybayev, M, Rametov, M, Nikolich, M, Hagius, S, Elzer, P, Blackburn, J. Brucellosis Transmission Between Humans and Domesticated Livestock In Southern Kazakhstan: Inferences Through MLVA Typing, UF Public Health and Health Professions Research Day, 2018

Waugh, S., Sytnik, I, Karibayev, T, Alimbayev, A, Ornybayev, M, Rametov, M, Nikolich, M, Hagius, S, Elzer, P, Blackburn, J. Brucellosis Transmission Between Humans and Domesticated Livestock In Southern Kazakhstan: Inferences Through MLVA Typing, AAG Annual Meeting, 2018

Waugh, S., Ball, J. Using statistical approaches to quantify the effects of ridesharing accessibility on Driving under the Influence (DUI) arrests in a university city, American Public Health Association Annual Meeting, 2016

Waugh, S., Varma, D., Striley, C., Cottler, L. Utilizing GIS to Visualize Hypertension Spread: A Comparative Study using HealthStreet Data, American Public Health Association Annual Meeting, 2015

Waugh, S., Varma, D., Striley, C., Cottler, L. Utilizing GIS to Visualize Hypertension Spread: A Comparative Study using HealthStreet Data, UF Public Health and Health Professions Research Day, 2015

Waugh, S. Geo-Spatial Risk Modeling for West Nile Virus in Tarrant County, TX Using Environmental and Demographic Data, AAG Annual Meeting, 2014